

Mongodb Practical Assignments

Exercise 1: Create a CollegeDB database and a Students collection. Insert 10 student documents with fields:

_id, Name, Age, Course, City, Email

Perform the following operations:

1. Display all students.
2. Display students from Surat.
3. Update the city of one student.
4. Update the course of all BCA students to BCA (AI).
5. Delete one student using _id.
6. Display the final list of students.

Exercise 2 : Create a CompanyDB database with an Employees collection. Insert 5 employee records with the following fields:

_id, Name, Department, Salary, Email, Experience (Years)

Perform these operations:

1. Display all employees.
2. Find employees with a salary greater than ₹50,000.
3. Increase the salary of employees in the HR department by updating their salary field.
4. Delete all employees with experience less than 2 years.
5. Display the remaining employee records.

Exercise 3 : Create a LibraryDB database and a Books collection. Insert 8 book documents containing:

Book ID, Title, Author, Category, Price, Available (true/false)

Perform the following:

1. Display all books.
2. Find all books in the Computer Science category.
3. Update the price of a selected book.
4. Mark one unavailable book as available.
5. Delete all books priced below ₹300.
6. Display the final collection.

Exercise 4 : Student–Department Relationship

Create two collections:

- students
- departments

Insert at least 5 students and 3 departments. Use a reference (department_id) and display student details along with department information using \$lookup.

Exercise 5 : Query Practice

Create an employees collection with fields:

- emp_id
- name
- salary
- department
- experience

Write queries to:

1. Display employees with salary greater than ₹50,000.
2. Display employees with experience between 3 and 8 years.
3. Display employees belonging to either the HR or IT department.
4. Sort employees by salary in descending order.
5. Display only the employee name and salary.

Exercise 6: Aggregation Practice

Create a sales collection with fields:

- product
- category
- price
- quantity

Write aggregation queries to:

1. Calculate the total sales amount for each category.
2. Find the average product price in each category.
3. Find the highest-priced product.
4. Count the number of products in each category.
5. Display the top 3 products with the highest sales value (price × quantity).